

MUHAMMAD ATIF ZAHEER

Department of Mathematics,
417 McAllister Building,
Penn State University,
State College, PA 16802
Email: mvz5421@psu.edu
Webpage: mazaheer.github.io

Research Interests

Analytic number theory, particularly moments of L-functions, zero-free regions, and pair correlation of zeros of derivatives of completed L -functions.

Education

- Ph.D. in Mathematics, Penn State University, 08/24 - 05/29 (expected)
Advisor: Amita Malik
- M.S. in Mathematics, Lahore University of Management Sciences, 09/22 - 05/24
Advisor: Emanuel Carneiro
Thesis Title: On Pair Correlation Conjecture and Bounds for the Averages of Montgomery's F -function
- B.S. in Mathematics, National University of Sciences and Technology, 08/18 - 05/22

Awards and Honors

- Jack and Eleanor Pettit Scholarship in Science, Eberly College of Science, Penn State University, 2024
- National Management Foundation Gold Medal, Lahore University of Management Sciences, 2024
- Dean's Honour List, Lahore University of Management Sciences, 2024
- Full Merit Scholarship, Lahore University of Management Sciences, 2022 - 2024
- High Achiever Award, National University of Sciences and Technology, 2022
- First Place - Pakistan Calculus Contest, IBA Sukkur University, 2022
- Second Place - National Undergraduate Mathematics Contest, Punjab University and National Mathematical Society of Pakistan, 2021
- Honor Roll, National University of Sciences and Technology, 2019

Papers and Preprints

- "Some Series Related to Extended Riemann Hypothesis for Dedekind Zeta Functions", preprint, arXiv:2512.19761, 2025

Teaching

Instructor, Penn State University

- Math 26: Plane Trigonometry, Spring 2026

- Math 26: Plane Trigonometry, Fall 2025

Teaching Assistant, Penn State University

- Math 26: Plane Trigonometry, Summer 2025
- Math 220: Matrices, Summer 2025
- Math 441: Matrix Algebra, Spring 2025
- Math 441: Matrix Algebra, Fall 2024

Teaching Assistant, Lahore University of Management Sciences

- Math 320: Algebra I, Spring 2024
- Math 230: Probability, Fall 2023
- Math 120: Linear Algebra, Summer 2023
- Math 205: Introduction to Analysis I, Spring 2023
- Math 101: Calculus I, Fall 2022

Talks

- “Littlewood’s Proof of Prime Number Theorem and the Theory of Almost Periodic Functions”, Number Theory Student Seminar, Penn State University, March 2026
- “ 10^{12} Equivalent Statements of Riemann Hypothesis”, Number Theory Student Seminar, Penn State University, November 2025
- “On Pair Correlation Conjecture and Bounds for the Averages of Montgomery’s F -function”, Thesis Defense, Lahore University of Management Sciences, May 2024
- “The Pair Correlation of the Zeros of the Riemann Zeta Function”, Graduate Student Seminar, Lahore University of Management Sciences, November 2023
- “Google Page Rank Algorithm”, Guest lecture for Math 120, Lahore University of Management Sciences, June 2023

Conferences and Workshops Attended

- Arithmetic Geometry at UNT 2, University of North Texas, Denton, August 2026
- International Congress of Mathematicians (ICM) 2026, Philadelphia, July 2026
- MPS Conference on Universal Statistics in Number Theory, Simons Foundation, New York, September 2025
- Midwest Arithmetic Geometry and Number Theory Series (MAGNTS), Ohio State University, March 2025
- The Legacy of John Tate, and Beyond, Harvard University, March 2025

Supervision

- Varnika Yadav & Kestian Apollos Orsetti, Directed Reading Project on Prime Number Theorem, Penn State University, Spring 2026

Professional Duties and Outreach

- Organizer, Number Theory Student Seminar, Penn State University, Fall 2025 & Spring 2026

- Facilitator, Bootcamp session on abstract algebra qualifying exam, Penn State University, August 2025
- Organizer, Salam Sessions for LUMS Students' Mathematical Society, 2023
- Instructor, Math Circles - LUMS STEM Bootcamp, Government Boys High School (Sukamaidan) Skardu, Gilgit Baltistan, Summer 2023

Miscellaneous

- Mentioned in the preface of fourth editions of texts "Analysis I" and "Analysis II" by Terence Tao
- Accepted and offered full funding for IAS/PCMI Graduate Summer School, July 2024 (did not attend)
- Accepted and offered full funding for Duke Number Theory School, May 2026 (did not attend)